🔷 High-level explanation (start with this)

This project builds an automated pipeline to generate high-quality synthetic images for object detection. The goal is to address limited or imbalanced real-world data by synthesizing new training images using real object crops and annotations.

🔷 Why synthetic data?

Manual annotation is expensive and time-consuming. Synthetic image generation allows us to increase dataset size, improve class balance, and expose models to diverse object placements without collecting new data.

🔷 End-to-end pipeline explanation (step-by-step, interview friendly)

1️⃣ Data ingestion from S3

The pipeline starts by downloading both annotated and unannotated sorted images from an S3 bucket. Duplicate downloads are avoided to ensure efficiency.

- Unannotated images → for background usage
- Annotated images → for object extraction

2️⃣ Annotation handling (CVAT → YOLO)

For annotated data, CVAT annotations in Datumaro JSON format are downloaded and combined into a single unified annotation file.

Two paths:

- Convert existing annotations to **YOLO format**
- Or **auto-annotate** using a pre-trained object detection model when annotations are missing

This ensures **consistent labeling** across the dataset.

3️⃣ Dataset consolidation

All images and YOLO labels are flattened into unified folders to standardize downstream processing.

This step simplifies:

- Crop generation
- Mask creation
- Synthetic placement

4️⃣ Object crop & mask generation

From the labeled dataset, object-level crops and corresponding binary masks are generated.

Outputs:

- Object crops
- Pixel-level masks
- Crop-level labels

These form the **building blocks** for synthetic images.

5️⃣ Synthetic image generation

Using the extracted object crops and masks, new images are synthesized by placing objects on background images.

Two strategies:

- **Spread-out placement** → realistic non-overlapping scenes
- **Clustered placement (optional)** → congested scenes with object repetition

This increases:

- Spatial diversity
- Object density variation
- Robustness of detection models

6️⃣ Phase-based synthetic generation (optional)

An optional phase-based generator creates images in multiple stages, gradually increasing object density and class balancing.

This helps:

- Control dataset distribution
- Target specific class counts
- Improve long-tail performance

7️⃣ Visualization & validation

Finally, synthetic images are visualized with bounding boxes overlaid from YOLO labels to verify annotation quality.

This acts as a **sanity check** before training.

🔷 Tech stack & reproducibility

The entire pipeline is Dockerized, ensuring reproducibility across environments. Dependencies include OpenCV, NumPy, Pandas, and AWS SDK (boto3) for seamless S3 integration.

🔷 One-line summary (very powerful)

I built a fully automated, Dockerized synthetic data generation pipeline that converts CVAT annotations into YOLO format, extracts object crops, and synthesizes diverse training images to improve object detection performance.

🔷 If interviewer asks "What impact does this have?"

You can say:

This pipeline significantly reduces annotation cost, improves class balance, and enables scalable dataset generation, leading to more robust object detection models in data-constrained environments.